it is a testing method where a tester investigates the internal structure, logic, design, or code of a software/application. It is commonly performed by developers who have a deep understanding of the source code to check which segment of code is not producing the desired output.

Advantages:

- Easier to find the type of data that can help in effectively testing the software.
- A tester can easily remove unnecessary lines of code that may cause errors in the software.
- Maximum bugs can be caught during the initial stages as a skilled tester or developer performs it.

Disadvantages:

- Expensive, as a tester should be skilled and aware of the coding techniques.
- Requires advanced tools such as debugger and code analyzers, hence difficult to maintain.

2. Black-box Testing

It is exactly opposite to the white-box test in which a tester evaluates the completeness of the software without looking into the code or internal design. In this testing, a tester does not know of the implementation details of the internal working of the system. It is also known as behavioural or input-output testing as the tester only focuses on the input fed and the output produced by the software. Black box testing is performed to validate functional requirements.

Advantages:

- No need to investigate the code and hence easier to perform.
- The moderately skilled tester can test the software.
- Focuses on inspecting outputs against the inputs. Hence it is performed from a user's point of view rather than a developer's point of view.

Disadvantages:

- As the tester does not know the internal code implementation of the software, this type of testing is not considered efficient.
- It is difficult to design the test cases.
- Code sections that can result in a defect cannot be targeted.